La coordinación

McCubbins-Rosenbluth

8-4-2026

Outline

1 SNTV

2 Clasificación del presupuesto

3 La regresión lineal múltiple

El sistema electoral japonés clásico

SNTV = single non-transferable vote

- $M > 1$ pero $V = 1$
- Los M candidatos más votados resultan electos (sistema de pluralidad con ganadores múltiples)
- Japón hasta 1998, Taiwán, Sudcorea, Afganistán (elección judicial mexicana en algunos casos)

Problema de coordinación del LDP

Caso $M = 4$ donde

partido A	tiene	100	electores fieles
B		24	
C		11	

Dos problemas

1. ¿cuántos nominas?
   - Si B manda uno, A puede ganar cuatro ssi coordinación perfecta de votantes
   - Si B manda dos, puede elegirlos ambos si A se concentra en alguno
   - ...
2. ¿cómo coordinas a tus votantes?

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Ministerios particularistas

EXPENDITURE ITEMS

GENERAL ACCOUNT

Prime minister's office

Science and Technology Agency

1. Promotion of research into peaceful uses for nuclear power
2. Science and Technology Agency research laboratory
3. Science and Technology promotion

Hokkaido Development Agency

4. Hokkaido road construction
5. Hokkaido fishing port facilities
6. Hokkaido housing construction industry

Ministry of Finance

Ministry of Finance (internal)

7. Supplement to People's Finance Corporation

Ministry of Education

Ministry of Education (internal)

8. National Treasury subsidies for educational expenses for the handicapped
9. Supplements to private schools

Ministry of Health and Welfare

10. Maintenance of environmental health and sanitation facilities
11. Support for the disabled

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Ministry of Health and Welfare
  10. Maintenance of environmental health and sanitation facilities
  11. Support for the disabled
  12. Welfare for the elderly
  13. Support for women
  14. Support for children

Ministry of Agriculture, Forestry, and Fisheries
  15. Fishing harbor facilities
  16. Strategic restructuring of agriculture
  17. Agricultural pensions
  18. Silk cultivation and horticulture promotion
  19. Diffusion of agriculture improvement
  20. Strengthening of wet rice agriculture
  21. Sugar price stabilization
  22. Strategic planning of food product distribution

Ministry of International Trade and Industry
- Ministry of International Trade and Industry (internal)
  23. Economic cooperation
  24. Promotion of industrial relocation
  25. Promotion of the computer industry
  26. Industrial plumbing facilities
- Agency of Industrial Science and Technology
  27. Promotion of mining and manufacturing technology
  28. Large-scale industrial research and development
  29. Agency of Industrial Science and Technology experimental research laboratory
- Agency of Natural Resources and Energy
  30. Measures for underground resources
- Small and Medium Enterprise Agency
  31. Measures for small and medium enterprises

Ministry of Transportation
- Ministry of Transportation (internal)
  32. Subsidies to account of Japan National Railway
  33. Subsidies to Japan National Railway Construction Corp.
  34. Subsidies for railroad track maintenance
  35. Coastline enterprises
  36. Subsidies for maritime transportation
  37. Stabilization of shipbuilding industry

Ministry of Posts and Telecommunications
  38. Integrated Telecommunications Research Lab
  39. Local Telecommunications Administration Bureau

Ministry of Labor
  40. Career change planning
  41. Labor Protection Office
  42. Employment Security Office

Ministry of Construction
- Ministry of Construction (internal)
  43. Strategic planning regarding crumbling of steep inclines and related expenses
  44. Coastline enterprises
  45. Housing construction
  46. City planning
  47. Work related to stream and river disasters
  48. Restoration of streams and rivers after disasters

Ministry of Home Affairs
- Ministry of Home Affairs (internal)
  49. Adjustment grants to cities, towns, and villages that are the sites of [defense] facilities

SPECIAL ACCOUNTS
  1. Foodstuff control
  2. Measures for the improvement of farm management
  3. Harbor improvement
  4. Airport improvement
  5. Road improvement

Ecuación

Para el ministerio particularista m en el año t

$$¥_{mt} = \beta_0 + \beta_1 \Delta ¥\text{resto}_{mt} + \beta_2 \,\%LDP_{t-1} + \ldots + \text{error}_i$$

Hipótesis

Todo lo demás constante, $\beta_2 > 0$

Estimación

Table 3.1 *Japanese budget regression*[a]

Independent variable	Estimated coefficient	Standard error	t-statistic
Constant	-0.63695	0.12224	-5.21082
Percentage change in budget remainder	1.24191	0.23015	5.39599
Percent LDP	1.04807	0.23272	4.50362
Lower House election	2.77452e-02	1.98918e-02	1.39480
Upper House election	1.94724e-02	2.27005e-02	0.85780
Ministry of Education	6.96370e-02	2.48803e-02	2.79888
Ministry of Posts & Telecom	-5.60421e-02	1.76490e-02	-3.17536

Note: Number of observations = 1,593
R-squared = 5.69691e-02 0.056

[a]Dependent variable: Partial Log of Real Budget for item i in year t.

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Ejemplo

$$Dvote_i = \beta_0 + \beta_1 \text{arrastre del gobernador}_i + \ldots + \text{error}_i$$

Estimaciones

Table 2 Four Models of Coattail Effects on Congressional Elections

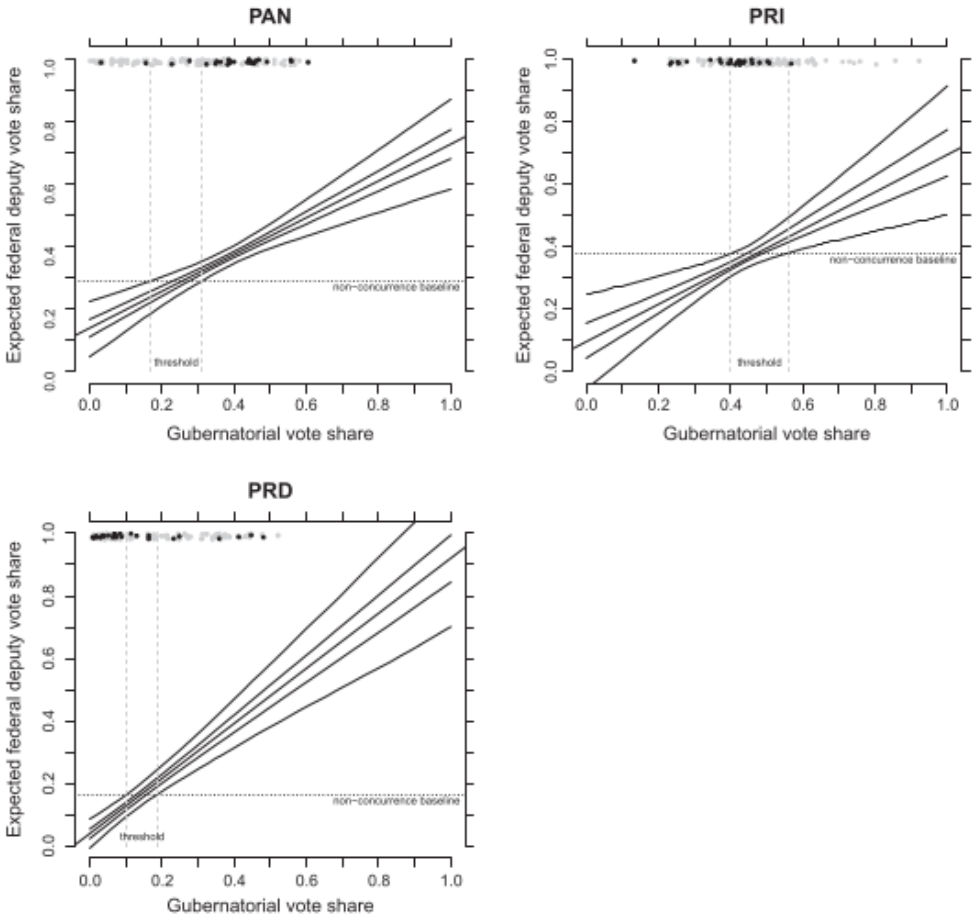
	1997–2009		1979–2009					
	(1) Shares		(2) Shares		(3) Shares		(4) Residuals	
Variable	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p	$\hat{\beta}$	p
Part A. PAN								
Constant	.135	.000	.084	.000	.074	.000	-.022	.004
RecentDvote	.586	.000	.671	.000	.787	.000		
GovOnlyConcurs	-.151	.005	-.057	.002			.017	.021
PresOnlyConcurs	-.097	.054	-.058	.011	-.047	.051	.042	.000
Gov&PresConcur	-.106	.102	-.063	.033			.038	.012
Gvote|GovOnlyConcurs	.584	.000	.405	.000			.760	.000
Gvote|Gov&PresConcur	.444	.043	.500	.002			.163	.325
Pvote|PresOnlyConcurs	.446	.002	.421	.000	.341	.000	.875	.000
Pvote|Gov&PresConcur	.018	.956	-.096	.638			.930	.000
IncumbentGovernor	.045	.021	.020	.155	.008	.583	.017	.123
IncumbentPresident	-.047	.004	-.025	.120	-.027	.128	.001	.926
Economy	-.145	.465	.204	.141	.206	.176	.110	.233
PartyCoalesced	.016	.416	-.001	.990	.007	.802	-.014	.382
N	160		352		352		352	
R^2	.82		.84		.81		.58	
Part B. PRI								
Constant	.187	.000	.092	.050	.062	.213	.017	.507
RecentDvote	.418	.000	.746	.000	.808	.000		
GovOnlyConcurs	-.288	.001	-.190	.000			-.019	.092
PresOnlyConcurs	-.249	.000	-.224	.000	-.199	.000	-.059	.005
Gov&PresConcur	-.235	.052	-.232	.001			-.045	.081
Gvote|GovOnlyConcurs	.620	.002	.291	.000			.737	.000
Gvote|Gov&PresConcur	.231	.478	.378	.270			.332	.185
Pvote|PresOnlyConcurs	.640	.000	.315	.000	.270	.005	.839	.000
Pvote|Gov&PresConcur	.362	.329	-.050	.876			.661	.013
IncumbentGovernor	.086	.000	.027	.168	.016	.449	.031	.062
IncumbentPresident	-.063	.006	-.033	.350	-.038	.298	-.014	.596
Economy	-.438	.005	.402	.090	.438	.087	.048	.815
PartyCoalesced	-.018	.132	.006	.801	.005	.834	-.003	.901
N	160		352		352		352	
R^2	.76		.88		.87		.56	

Estimación

	1997–2009	
	(1) Shares	
Variable	$\hat{\beta}$	p
Part A. PAN		
Constant	.135	.000
RecentDvote	.586	.000
GovOnlyConcurs	-.151	.005
PresOnlyConcurs	-.097	.054
Gov&PresConcur	-.106	.102
Gvote|GovOnlyConcurs	.584	.000
Gvote|Gov&PresConcur	.444	.043
Pvote|PresOnlyConcurs	.446	.002
Pvote|Gov&PresConcur	.018	.956
IncumbentGovernor	.045	.021
IncumbentPresident	-.047	.004
Economy	-.145	.465
PartyCoalesced	.016	.416
N	160	
R^2	.82	

Simulaciones

FIGURE 3 Gubernatorial Coattails in Congressional Elections 1997–2009



Note: Plots prepared with MCMC estimates of model 1 in Table 2. Panels report the median, 50%, and 95% intervals of the posterior distribution. The simulation scenario for each party has the following features: the last deputy vote share is set at the party's median value for the period; there is a concurrent gubernatorial but not presidential election; the party has no incumbent governor nor president; the state's economic performance set at the median value for the period; and no party coalesced in the congressional election. The non-concurrence baseline is the median of the posterior distribution when *GovOnlyConcurs* = 0 in the same scenario. Dots are the party's actual gubernatorial vote shares (y-jittered for visibility), heavier points for races concurrent with congressional elections.